

Lecture 11

Some Special Continuous Probability Distributions

04.05.2026

Outline

- 1 Uniform distribution
- 2 Exponential distribution
- 3 Normal distribution

Uniform Distribution

Let's define a continuous probability distribution that has some constant value c between α and β where $\alpha < \beta$. What is the pdf?

$$f(x) = \begin{cases} c & \text{if } \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$

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Definition

A continuous random variable X has a Uniform distribution on the interval (α, β) if its pdf is

$$X \sim \text{Unif}(\alpha, \beta) \iff f(x) = \frac{1}{\beta - \alpha} \cdot \mathbf{1}_{(\alpha, \beta)}(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$

Cdf of Uniform distribution

For any $x \in (\alpha, \beta)$,

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{\alpha}^x \frac{1}{\beta - \alpha} dt = \frac{x - \alpha}{\beta - \alpha}$$

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- $X \sim \text{Unif}(\alpha, \beta)$, then its cdf is
$$F(x) = \begin{cases} 0 & \text{if } x \leq \alpha \\ \frac{x - \alpha}{\beta - \alpha} & \text{if } \alpha < x < \beta \\ 1 & \text{if } x \geq \beta \end{cases}$$

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$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 = \frac{\beta^2 + \beta\alpha + \alpha^2}{3} - \frac{(\beta + \alpha)^2}{4} \\ &= \frac{4\beta^2 + 4\beta\alpha + 4\alpha^2}{12} - \frac{3\beta^2 + 6\alpha\beta + 3\alpha^2}{12} \\ &= \frac{\beta^2 - 2\beta\alpha + \alpha^2}{12} = \frac{(\beta - \alpha)^2}{12} \end{aligned}$$

Uniform distribution: probability calculation

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Let X denote the time I arrive the bus stop.

$$X \sim \text{Unif}(0, 60)$$

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The bus arrives at time $t = 0, 10, 20, 30, 40, 50, 60$.

So the intervals of my arriving time such that I will wait more than 7 min:

$$B = (0, 3) \cup (10, 13) \cup (20, 23) \cup (30, 33) \cup (40, 43) \cup (50, 53)$$

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$$B = (0, 3) \cup (10, 13) \cup (20, 23) \cup (30, 33) \cup (40, 43) \cup (50, 53)$$

$$P(X \in B) = (3 \times 6)/60 = 0.3$$

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- 2 Exponential distribution**
- 3 Normal distribution

Exponential distribution

Definition

A continuous random variable X has a Exponential distribution distribution with $\lambda > 0$ if its pdf is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

- $X \sim \text{Exp}(\lambda)$

Exponential distribution

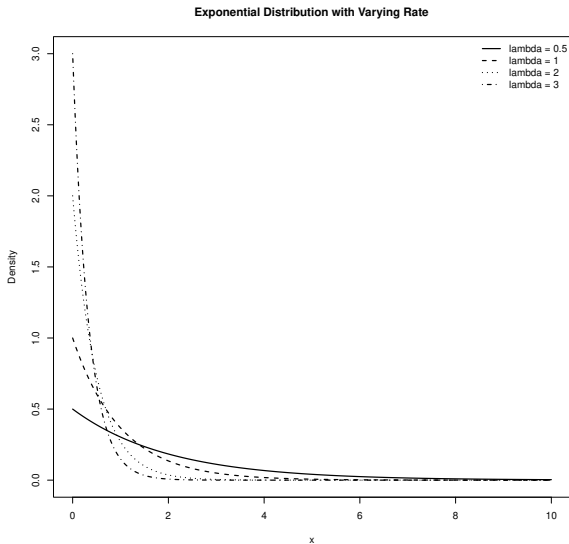
Definition

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$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

- $X \sim \text{Exp}(\lambda)$
- $F(x) = 1 - e^{-\lambda x}$ if $x \geq 0$ and 0 elsewhere.
- $E(X) = \frac{1}{\lambda}$
- $\text{Var}(X) = 1/\lambda^2$

PDF plot



Example

Suppose that the length of a phone call in minutes is an exponential random variable with parameter $\lambda = 1/10$. If someone arrives immediately ahead of you at a public telephone booth, find the probability that you will have to wait

- Ⓐ more than 10 minutes;
- Ⓑ between 10 and 20 minutes.

Example

Let X be an exponentially distributed random variable with $\lambda = 2$. Find the value of c for which $P(X > c) = .25$.

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$$X \sim N(\mu, \sigma^2) \iff f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ where } x \in \mathbb{R}$$

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Pdf: unimodal and symmetric, bell shaped continuous random variable

The normal pdf is well-defined

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

Use the fact that normal pdf is well define, calculate the integral

$$\int_{-\infty}^{\infty} e^{-\frac{(x-1)^2}{10}} dx = ?$$

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Suppose we have a random variable $X \sim N(\mu = 1, \sigma^2 = 5)$, then

$$1 = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi \cdot 5}} e^{-\frac{(x-1)^2}{10}} dx$$

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Mean and variance of Normal random variable

If $X \sim N(\mu, \sigma^2)$, and $Z = \frac{X-\mu}{\sigma}$, then Z has a standard normal distribution (more later in Chapter 5.7).

$$Z \sim N(0, 1)$$

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$$\begin{aligned} E[Z] &= \int_{-\infty}^{\infty} x f_Z(x) dx = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-\frac{x^2}{2}} dx \\ &= -\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \Big|_{-\infty}^{\infty} = 0 \end{aligned}$$

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$$\begin{aligned} Var[Z] &= E[Z^2] - (E[Z])^2 = E[Z^2] = \int_{-\infty}^{\infty} x^2 f_Z(x) dx = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}} dx \quad (\text{integration by parts : } u = x, dv = x e^{-x^2/2} dx) \\ &= \frac{1}{\sqrt{2\pi}} \left\{ -x e^{-\frac{x^2}{2}} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx \right\} \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = 1 \end{aligned}$$

68-95-99.7 Rule

- A random variable X has a normal distribution,
 - ▶ about 68% probability X falls within 1 SD of the mean,
 - ▶ about 95% probability X falls within 2 SD of the mean,
 - ▶ about 99.7% probability X falls within 3 SD of the mean.
- The probability of X falls 4, 5, or more standard deviations away from the mean is very low.

Normal probability calculation

- We denote $\phi(x)$ and $\Phi(x)$ as pdf and cdf of the standard normal distribution respectively.

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- We denote $\phi(x)$ and $\Phi(x)$ as pdf and cdf of the standard normal distribution respectively.
- Probability calculations for X in terms of Z :

$$\begin{aligned} P(a < X < b) &= P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right) \end{aligned}$$

Example

If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, find
(a) $P(2 < X < 5)$; (b) $P(X > 0)$; (c) $P(|X - 3| > 6)$.

Example

Suppose that X is a normally distributed random variable with mean μ and variance σ^2 . If $P(X < 10) = 0.67$ and $P(X < 20) = 0.975$, approximate μ and σ^2 .

At Heinz ketchup factory the amounts which go into bottles of ketchup are supposed to be normally distributed with mean 36 oz. and standard deviation 0.11 oz. Once every 30 minutes a bottle is selected from the production line, and its contents are noted precisely. If the amount of ketchup in the bottle is below 35.8 oz. or above 36.2 oz., then the bottle will fail the quality control inspection. What's the probability that the amount of ketchup in a randomly selected bottle is less than 35.8 ounces?

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$$z = \frac{x - \mu}{\sigma} = \frac{35.8 - 36}{0.11} = -1.82$$

$$P(X < 35.8) = P(Z < -1.82) = 0.0344$$

Normal approximation to the Binomial distribution

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Let $X \sim \text{Bin}(n, p)$. When n is large enough, or more specifically, $np(1 - p) \geq 10$, the binomial distribution can be approximated by the normal distribution

$$P(X = i) \approx P(i - 0.5 < Y < i + 0.5), \quad Y \sim \text{N}(\mu, \sigma^2)$$

with parameters $\mu = np$ and $\sigma^2 = np(1 - p)$.

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with parameters $\mu = np$ and $\sigma^2 = np(1-p)$.

- Probability calculation (actually, approximation)

$$\begin{aligned} P(a \leq X \leq b) &\approx P(a - 0.5 < Y < b + 0.5) \\ &= P\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}} < \frac{Y - \mu}{\sigma} < \frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right) \\ &= \Phi\left(\frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}}\right) \end{aligned}$$

The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

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- We are given that $X \sim \text{Bin}(n = 450, p = 0.3)$, and we are asked for the probability

$$P(X > 150) = p(151) + \cdots + p(450) = 1 - p(0) - p(1) - \cdots - p(151)$$

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- To use normal approximation, first check conditions

$$np(1 - p) = 450 \times 0.3 \times 0.7 = 94.5 \geq 10$$

Use Normal approximation with continuity correction,

$$\begin{aligned} P(X > 150) &\approx P(X > 150 + 0.5) \\ &= P\left(Z > \frac{150.5 - 450(0.3)}{\sqrt{450(0.3)(0.7)}}\right) \\ &= P(Z > 1.59) \\ &= 1 - 0.9441 = 0.0559 \end{aligned}$$

Recap: Normal distribution $X \sim \mathcal{N}(\mu, \sigma^2)$

- Mean μ , variance σ^2
- Symmetry

$$f(\mu - x) = f(\mu + x), F(\mu - x) = 1 - F(\mu + x)$$

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$$X \sim N(\mu, \sigma^2) \iff Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

- Find probability using $\Phi(\cdot)$ table

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- Normal approximation to Binomial $X \sim \text{Bin}(n, p)$

$$Y \sim \mathcal{N}(\mu = np, \sigma^2 = np(1 - p))$$

$$P(X = i) \approx P(i - 0.5 < Y < i + 0.5)$$

Review: continuous distributions

Name	Range	pdf $f(x)$	mean	variance
$\text{Unif}(\alpha, \beta)$	$[\alpha, \beta]$	$\frac{1}{\beta - \alpha}$	$\frac{\alpha + \beta}{2}$	$\frac{(\beta - \alpha)^2}{12}$
$N(\mu, \sigma^2)$	$(-\infty, \infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$	μ	σ^2